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| <b>3.3 Graphs</b> | <p>plot and draw graphs with equation:<br/> <math>y = Ax^3 + Bx^2 + Cx + D</math> in which:</p> <p>(i) the constants are integers and some could be zero</p> <p>(ii) the letters <math>x</math> and <math>y</math> can be replaced with any other two letters</p> <p>or:</p> <p><math>y = Ax^3 + Bx^2 + Cx + D + E/x + F/x^2</math></p> <p>in which:</p> <p>(i) the constants are numerical and at least three of them are zero</p> <p>(ii) the letters <math>x</math> and <math>y</math> can be replaced with any other two letters</p> <p>find the gradients of non-linear graphs</p> | <p> <math>y = x^3</math>,<br/> <math>y = 3x^3 - 2x^2 + 5x - 4</math>,<br/> <math>y = 2x^3 - 6x + 2</math>,<br/> <math>V = 60w(60 - w)</math> </p> <p> <math>y = \frac{1}{x}, x \neq 0</math>,<br/> <math>y = 2x^2 + 3x + 1/x</math>,<br/> <math>x \neq 0</math>,<br/> <math>y = \frac{1}{x}(3x^2 - 5)</math>,<br/> <math>x \neq 0</math>,<br/> <math>W = \frac{5}{d^2}, d \neq 0</math> </p> <p>By drawing a tangent</p> |
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|  | <p>find the intersection points of two graphs, one linear (<math>y_1</math>) and one non-linear (<math>y_2</math>), and recognise that the solutions correspond to the solutions of <math>y_2 - y_1 = 0</math></p> <p>calculate the gradient of a straight line given the coordinates of two points</p> <p>recognise that equations of the form <math>y = mx + c</math> are straight line graphs with gradient <math>m</math> and intercept on the <math>y</math> axis at the point <math>(0, c)</math></p> <p>find the equation of a straight line parallel to a given line</p> | <p>The <math>x</math>-values of the intersection of the two graphs:</p> $y = 2x + 1$ $y = x^2 + 3x - 2$ <p>are the solutions of:</p> $x^2 + x - 3 = 0$ <p>Similarly, the <math>x</math>-values of the intersection of the two graphs:</p> $y = 5$ $y = x^3 - 3x^2 + 7$ <p>are the solutions of:</p> $x^3 - 3x^2 + 2 = 0$ <p>Find the equation of the straight line through <math>(1, 7)</math> and <math>(2, 9)</math></p> |
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